

JAD FALAQ

Generative AI & AI Agents Engineer – LLMs, RAG & Multi-agent Systems

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6-month final-year engineering internship starting January 2027

PROFILE

Generative AI Engineer, final-year AI engineering student at **ENSIAS**, with internships at **Renault Digital** and **AKWA Group**. Designed and **shipped to production a multi-agent LLM system with RAG** adopted by a QA team, from a Copilot Studio prototype to a full-stack application, and built **cvmatch**, a LangGraph agentic tool **deployed online**. Specialized in **AI agents**, **hybrid RAG** and **reliable LLM outputs**, following the principle **"the LLM proposes, the code verifies"** with a **human in the loop**.

PROFESSIONAL EXPERIENCE

AI Engineer Intern – Quality Engineering

Renault Digital | Internship | Casablanca, Morocco — Jun. 2026 – Sep. 2026

- **Multi-agent AI assistant in production.** Engineers describe their need in text, PDF or image, the assistant asks clarifying questions and proposes a complete Jira ticket (acceptance criteria, Gherkin scenarios), created only after validation. Parallel agents (analysis, test design, Jira integration), Copilot Studio prototype then full stack (FastAPI, React/TypeScript, Microsoft Copilot API). Writing time cut from 10–20 min to 1–2 min per ticket.
- **RAG & guardrails.** RAG on documentation and past tickets (OCR, structure-based chunking, pgvector, MMR) for consistency with team history, sensitive-data masking (Presidio, spaCy), similarity threshold with fallback on model failure, outputs validated by Pydantic and the Cucumber parser, 85% of generations accepted without edits.
- **Automation & Agile.** Four Power Automate workflows launched on my own initiative (Smoke and TNR tracking files, automatic reminders, duplicate handling), one-week sprints in a team of about ten full-stack and DevOps engineers.

ML & Data Engineer – IT Data Team

AKWA Group | Internship | Casablanca, Morocco — Jun. 2025 – Sep. 2025

- **Missing transaction detection.** End-to-end ML pipeline (CRISP-DM) on accounting entries from Afriware, the Odoo and JDE ERPs and internal files (6M+ transactions), comparing supervised and unsupervised models, Random Forest selected (98% accuracy, 85% F1-score) and integrated into the existing system.

TECHNICAL SKILLS

- **LLMs & agents.** LangGraph, LangChain, multi-agent systems, tool calling, human-in-the-loop, prompt engineering, structured outputs (JSON, Pydantic), Azure OpenAI, OpenAI, Anthropic, Gemini, Mistral, Ollama
- **RAG.** Embeddings, pgvector, ChromaDB, semantic and agentic chunking, hybrid search (BM25), MMR, RRF, reranking, OCR
- **Machine & deep learning.** PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, fine-tuning, CNN, LSTM, YOLOv8, U-Net, GNN, knowledge distillation
- **Evaluation & reliability.** LLM-as-a-judge, Langfuse, LangSmith, LLM benchmarks, threshold calibration, precision, recall, F1, mAP
- **MLOps & deployment.** Docker, GitHub Actions (CI/CD), MLflow, DVC, FastAPI, Azure (Container Apps), pytest, Git, Linux
- **Development & automation.** Python, SQL, TypeScript, React, Next.js, PostgreSQL, Power Automate, Agile

EDUCATION

State Engineering Degree – Artificial Intelligence Engineering (2IA)

ENSIAS, Rabat | Sep. 2024 – Jun. 2027

- Machine learning and deep learning (neural networks, reinforcement learning), statistical and probabilistic learning
- Big Data and distributed systems, data engineering, time series, multimedia processing (text, image)

Preparatory Classes for the French Grandes Écoles (CPGE) – Mathematics & Physics

CPGE Mohammed V, Casablanca | Sep. 2022 – Jun. 2024

ACADEMIC PROJECTS

- **cvmatch – LangGraph agent & hybrid RAG (live demo).** Compares a CV with a job offer skill by skill, retrieves evidence through hybrid RAG (BM25 + embeddings, RRF fusion), computes a score in code and suggests rewrites without inventing anything. "The LLM proposes, the code verifies", with word-for-word citation checks, human validation and Docker on Azure
- **RAG strategies.** RAG pipeline (Azure OpenAI, ChromaDB) comparing each step, from chunking (character, semantic, LLM-based) to retrieval (similarity, threshold, MMR, multi-query) and RRF fusion, with multimodal RAG
- **Dental pathology detection – Computer vision & MLOps.** YOLOv8 on 1,808 X-rays (24,891 audited annotations, 14 classes remapped into 7), evaluated on validation, test and external sets, with a continuously deployed pipeline (DVC, MLflow, CI/CD, Azure)
- **NLP & fine-tuning.** Fine-tuned AraBERT to detect medical misinformation in Arabic tweets, 87% accuracy and 0.87 macro F1

SOFT SKILLS

- **Entrepreneurial mindset.** Automations launched on my own initiative, e-commerce websites built for clients
- **Listening & ownership.** Product shaped by user feedback, no AI action without human validation

LANGUAGES

French (native) – Arabic (native) – English (bilingual)